Table 26. Embedded reset and power control block characteristics

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
$t_{RSTTEMPO}$ (1)	POR temporization when V_{DD} crosses V_{POR}	V_{DD} rising	-	270	500	µs
V_{POR} (1)	Power-on reset threshold	-	1.9	1.94	1.98	V
V_{PDR} (1)	Power-down reset threshold	-	1.88	1.92	1.96	V
V_{BOR1}	Brownout reset threshold 1	V_{DD} rising	2.05	2.10	2.18	V
		V_{DD} falling	1.95	2.00	2.08	
V_{BOR2}	Brownout reset threshold 2	V_{DD} rising	2.20	2.31	2.38	V
		V_{DD} falling	2.10	2.21	2.28	
V_{BOR3}	Brownout reset threshold 3	V_{DD} rising	2.50	2.62	2.68	V
		V_{DD} falling	2.40	2.52	2.58	
V_{BOR4}	Brownout reset threshold 4	V_{DD} rising	2.80	2.91	3.00	V
		V_{DD} falling	2.70	2.81	2.90	
$V_{hyst\_POR\_PDR}$	Hysteresis of V_{POR} and V_{PDR}	-	-	20	-	mV
V_{hyst_BOR}	Hysteresis of V_{BORx}	-	-	100	-	mV
$I_{DD(BOR)}$ (1)	BOR consumption from V_{DD}	-	-	2.2	2.5	µA

1. Specified by design. Not tested in production.

5.3.4 Embedded voltage reference

The parameters given in *Table 27* are derived from tests performed under the ambient temperature and supply voltage conditions summarized in *Table 24: General operating conditions*.

Table 27. Embedded internal voltage reference

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
V_{REFINT}	Internal reference voltage	-	1.182	1.212	1.232	V
$t_{S\_vrefint}$ (1)(2)	ADC sampling time when reading the internal reference voltage	-	4	-	-	µs
$t_{start\_vrefint}$ (2)	Start time of reference voltage buffer when ADC is enable	-	-	8	12	µs
$I_{DD(VREFINTBUF)}$ (2)	V_{REFINT} buffer consumption from V_{DD} when converted by ADC	-	9	13.5	23	µA
ΔV_{REFINT} (2)	Internal reference voltage spread over the temperature range	V_{DD} = 3 V	-	30	50	mV
T_{Coeff}	Averange temperature coefficient	-	-	20	70	ppm/°C
A_{Coeff}	Long term stability	1000 hours, T = 25 °C	-	300	1000	ppm
$V_{DDCoeff}$	Voltage coefficient	3.0 V < V_{DD} < 3.6 V	-	250	1200	ppm/V

1. The shortest sampling time can be determined in the application by multiple iterations.
2. Specified by design. Not tested in production.